

Gil Lopes Bueno · Senior Blockchain Engineer · Sao Paulo, Brazil (UTC-3)

gilbueno.mail@gmail.com github.com/melanke linkedin.com/in/gilbueno https://gil.solutions +55 11970629099

I'm a Web3 Software Engineer with 8+ years building decentralized systems for mainstream users. I've led self-custodial wallets, abstract accounts with sponsored transactions, and multi-chain SDKs used across production apps. On Neon Wallet, that work backed over \$1 billion in traded volume.

On the protocol side I've designed DeFi systems end to end, from bonding-curve launchpads to Uniswap v4 hooks, plus prediction markets with vault-based yield. I worked as main developer at a firm that started as a security auditing shop, and it taught me to treat security as an architecture decision, not a final review step.

Recently I've been bringing AI into this work, integrating LLM agents into development and auditing workflows and building tooling that generates smart contracts from natural language. With 19 years in fullstack and 12 leading teams of up to 30, I care about making blockchain feel as smooth as a normal web app.

Notable Achievements

- Neon Wallet – US\$1B+ in traded volume
- iTrack – 50M invoices; 60k couriers; 2k companies
- Sharity – R\$2M in donations; 100k users
- Multilaser Runin – Embedded in 20M+ devices
- Apptite – US\$1.6M GMV; 100k orders; 50k users
- Desabafa – 700k posts

Skills

BLOCKCHAIN: Ethereum (2020-now), Solidity (2020-now), Hardhat (2020-now), Foundry (2025-now), Protocol Architecture (2023-now), Fuzzing (2023-now), Gas Optimization (2023-now), Wagmi (2023-now), Viem (2024-now), The Graph (2023-2025), Solana (2021-2025), Flow (2022-2023), Neo N3 (2021-2024), VM Compiler Development (2018-2024), Wallet Development (2021-2025), NFT (2018-2023), Crypto Currency (2018-2024), DEX (2021-now), AMM (2023-now), Ethers (2021-2025), Audit Prep (2023-now), Slither (2023-now)

AI ENGINEERING: AI Process Automation (2025-now), Agent Development (2025-now), MCP (2025-now), Claude Skills (2025-now), Harness Engineering (2026-now), Spec-Driven Development (2025-now), Claude (2025-now), Anthropic API (2025-now), OpenAI API (2023-now), Claude Code (2025-now), Cursor (2024-now), Context Engineering (2025-now), Agentic Workflows (2025-now), Evals (2025-now), RAG (2024-now)

BACKEND: Node.js (2012-2025), Java (2008-2024), Kotlin (2016-2024), MySQL (2013-2024), PostgreSQL (2013-2025), NoSQL (2018-now), Prisma (2021-2025), GraphQL (2021-2024), REST (2012-now), WebSockets (2016-2025), Microservices (2019-now), Distributed Systems (2019-now), Solution Architecture (2016-2025), Redis (2015-2024), Docker (2016-now), AWS (2013-2025), ECS (2016-2025), S3 (2015-2025), SNS (2015-2025), SQS (2015-2025), Terraform (2019-2025), CI/CD (2018-now), GitHub Actions (2020-now), C# (2018-2024), Python (2018-2024), Express (2012-now), TypeGraphQL (2021-2024), Apollo (2021-2024), Jersey (2020-2023), PayPal (2016-2020), Elasticsearch (2019-2025)

WEB FRONTEND: JavaScript (2008-now), TypeScript (2016-now), React.js (2016-now), Next.js (2021-now), Tailwind (2020-2025), SvelteKit (2023-2024), Chakra UI (2020-2023), React Query (2021-2025), Redux Toolkit (2021-2024), ECharts (2019-2025), Valtio (2023-2025), Vite (2022-2025), Jest (2018-2025), Playwright (2023-2024), Storybook (2020-2024), React Hook Form (2020-2023)

LANGUAGES: English (C2), Portuguese (Native)

Work Experience

33Labs • Software Engineer | Main Smart Contract Developer • Sep 2025 - current

At 33Labs (formerly 33Audits), I've worked on the design and development of advanced DeFi protocols, focusing on composability, capital efficiency, and developer experience.

The company began as a smart contract auditing firm, which has given me hands-on, daily experience identifying and preventing vulnerabilities. I've developed protocols end to end — architecture, implementation, and testing — with security and gas optimization as constant priorities. That led me to a key conviction: the auditing process should start at the architecture stage, before any code is written. To support it, I've integrated specialized tooling and AI agents into the development and auditing workflow.

Tech: Solidity, Foundry, Ethereum, EVM, Uniswap V3, Uniswap V4, System Architecture, AI Tooling, REST · <https://www.33labs.ai/>

33Labs • American Spend • Software Engineer | Main Smart Contract Developer • Mar 2026 - May 2026

American Spend is a prediction market protocol combining yield generation with market-based price discovery through a hybrid market structure.

Designed a hybrid market model starting with a parimutuel phase and transitioning into a Central Limit Order Book (CLOB).

Integrated vault-based yield strategies to generate returns on idle capital during market activity.

Led the end-to-end design and implementation of the smart contract architecture, ensuring seamless transitions between market phases without disrupting user positions or incentives.

Drove close collaboration with auditors throughout development, incorporating feedback early to strengthen security and reduce iteration cycles.

Tech: Solidity, Foundry, Prediction Market, Vault, CLOB, Ethereum, EVM · <https://spend.market/>

33Labs • Mosaic • Software Engineer | Main Smart Contract Developer • Sep 2025 - Feb 2026

Mosaic is a protocol designed to make smart contract development more accessible, cost-efficient, and secure through flexible, pre-audited, and composable contracts, with AI-assisted tooling for generating contract compositions from natural language.

Contributed to the core architecture and designed reusable protocol modules with a strong focus on composability.

Designed and implemented a Launchpad workflow powered by a bonding curve, graduating into a Uniswap v3 liquidity pool for smooth price discovery and liquidity transition.

Built a Uniswap v4 hook combining CLMM behavior with a bonding curve model, later transitioning into a standard AMM.

Developed abstractions to support modular and extensible DeFi primitives.

Tech: Solidity, Foundry, Uniswap V3, Uniswap V4, Ethereum, EVM · <https://mosaic.build/>

Jodobix • Software Engineer | Creator • Mar 2025 - Jun 2025

Jodobix is a decentralized betting game designed to be fully fair and autonomous. It leverages blockchain technology to eliminate intermediaries and guarantees that all betting value is distributed among players and contributors. I developed the entire project on my own, including the design of secure random number generation strategies that do not rely on trusted third parties.

Tech: Solidity, Hardhat, Optimism, TypeScript, React.js, Next.js, Ethereum, EVM, Tailwind, React Query, Valtio, Ethers, The Graph, Solana · <https://jodobix.com> · Contract on Optimism

BuidlGuidl Batch Program • Mentor • Feb 2025 - Jan 2026

BuidlGuidl is one of the most impactful builder communities in the Ethereum ecosystem. One of its core initiatives is the Batch Program, which helps onboard new developers into the Ethereum space. As a mentor in the program, I guide participants through their first steps contributing to open source using Solidity, Hardhat, Wagmi, and Next.js.

Tech: Solidity, Hardhat, TypeScript, React.js, Next.js, Ethereum, EVM, Wagmi, Viem · <https://buidlguidl.com/batches>

Simpli • Co-founder | Software Engineer | CTO • Oct 2013 - May 2025

I co-founded Simpli as a two-person startup focused on building a B2C mobile product, which quickly evolved into a fast-growing software house serving a wide range of clients. In its first year, the company pivoted to delivering custom distributed applications and scaled organically through consistent delivery and client satisfaction. Over 11 years, we delivered 50+ successful digital products for both startups and enterprise clients, while also launching and maintaining our own proprietary platforms.

From the beginning, Simpli was an early adopter of emerging technologies like mobile development and blockchain. I played a key role in shaping both the technical direction and business strategy of the company, leading system architecture, technical roadmaps, and documentation, and building an engineering org of 30 including 5 team leads. My work ranged from hands-on technical leadership to driving innovation through research, process design, and technology adoption — helping turn product ideas into real businesses by aligning technical execution with market opportunities.

Provisioning AWS infrastructure was a recurring part of that work. With 50+ projects, we needed a standard: a checklist for spinning up a new account, a baseline set of IAM roles, and a reusable Terraform starter kit so no project's infra started from a blank page — the same template became a paved road that sped up every project's first deploy. Cost optimization started reactively, after a client flagged a high AWS bill, but past a certain point we set up anomaly-alert triggers instead of waiting for the next complaint.

Tech: Java, Kotlin, C#, TypeScript, Node.js, React.js, React Native, Next.js, Angular, SvelteKit, Android, iOS, Xamarin, Electron.js, JQuery, Backbone, Python, R, MySQL, PostgreSQL, Redis, GraphQL, Jersey, AWS, ECS, S3, SNS, SQS, Web3, Blockchain, Smart Contracts, Cryptography, Cadence, Flow, Prisma, WebSockets, PayPal, ElasticSearch, REST, JavaScript, Tailwind, React Query, Redux Toolkit, ECharts, Valtio

Simpli • Enclave Wallet • Software Engineer | Product Owner | UI/UX Designer • Jun 2024 - Feb 2025

Enclave is a Wallet built to embrace non-blockchain users. It combines cutting edge technologies like Abstract Accounts, WebAuthn and Gasless transactions, to allow the onboarding to be smooth as traditional web applications. Working with a small

team, I was responsible for the product vision, usability, development of the whole wallet frontend and contributing to the Smart Contracts.

Tech: TypeScript, React.js, Next.js, Node.js, PostgreSQL, AWS, ECS, S3, SNS, SQS, Smart Contracts, Cryptography, Prisma, WebSockets, ElasticSearch, REST, Tailwind, React Query, ECharts, Valtio · <https://enclavewallet.com>

Simpli · Blockchain Services Library • Software Engineer | Techlead • Aug 2023 - Jul 2024

BSLib is a multi-chain library designed to perform common wallet operations in a generic manner, abstracting and normalizing the unique characteristics of each blockchain. It includes implementations for NeoN3, NeoLegacy, and various EVM networks. This library is extensively used by Neon Wallet Desktop and Mobile applications. As the creator of BSLib, I aimed to enable code reuse across the wallets maintained by my team.

Tech: TypeScript, Node.js, Blockchain, Cryptography, REST · <https://github.com/CityOfZion/blockchain-services>

Simpli · Letter • Software Engineer | TechLead • Feb 2022 - Apr 2023

In partnership with the Associated Press, Dapper Labs, and COZ, Letter is a multi-chain (Neo and Flow) platform that provides authentication mechanisms based on NFTs. This allows systems to validate access in a decentralized manner. I was responsible for architecting the solution, which included SmartContracts on both networks, an SDK that integrates both networks simultaneously, and several key integrations.

Tech: TypeScript, React.js, Smart Contracts, Cadence, Flow, REST, Chakra UI, React Query, Redux Toolkit

Simpli · Neon Wallet • Software Engineer | TechLead • July 2021 - July 2024

Neon is the leading wallet in the Neo ecosystem, with over \$1 billion in traded volume. I was responsible for architecting its mobile version and later contributed to the desktop app. During my time on the project, I tackled key challenges such as supporting multiple blockchain networks, managing multiple accounts simultaneously, implementing WalletConnect integration, and developing the protocol for network interaction, along with several other critical integrations.

Tech: TypeScript, React.js, React Native, Blockchain, Electron.js, REST, React Query, Redux Toolkit · <https://coz.io/neon-wallet/>

Simpli · LDC's She Digital • Software Engineer | TechLead • Oct 2019 - Nov 2020

Louis Dreyfus Company, one of the largest commodity traders in the world, commissioned a 'Safety, Health, and Environment' management platform for use across all its global units — each with its own local regulations, forms, and reporting requirements. I owned it end to end: architecture, development, and integration with Azure Active Directory for authentication and user management. The core challenge was translating that sprawl of per-unit requirements into one flexible application instead of one branch per country, so I modeled forms, fields, and workflows as configurable data instead of hardcoded screens — letting a compliance team in a new unit stand up its own SHE process by configuring the platform, not by requesting a code change and a deploy.

Tech: TypeScript, React.js, Kotlin, Java, MySQL, REST, AWS, ECS, ECharts

Simpli · iTrack • Software Engineer | TechLead • Nov 2016 - 2024

I was the lead engineer who built iTrack Brasil from the ground up, owning its system design through its first two years and scaling it into a B2B delivery platform integrating multiple systems, with nearly 60,000 couriers and over 50 million invoices processed across 2,000 registered companies — growth that led to its acquisition by MadeiraMadeira in 2021. Simpli carried it through that acquisition and ran it three more years before handing it over in 2024, with me on it as CTO, coming back alongside the team when the work needed the depth. One of those returns: a billing alarm fired, Cost Explorer put the spike on Postgres at fiscal month-end, and Performance Insights traced it to clients posting invoices one per request instead of in batch — a transaction, and a WAL flush, per invoice, which no instance class fixes. Ingestion moved behind an SQS queue with concurrent workers committing in batches, autoscaled on backlog per worker, and that fix exposed a second write load underneath: courier position pings, which moved to Redis for last-known position with Parquet history on S3.

Tech: TypeScript, React.js, Kotlin, Java, AWS, S3, SQS, Redis, PostgreSQL, ECS, REST · <https://itrackbrasil.com.br>

SIMET - NIC.br • Software Engineer • 2010 - 2013

Brazil's official internet quality measurement tool. I led its migration off Java Applets to plain JavaScript.

<https://simet.nic.br>

FingerPrint, Escala, NetStartup • Software Engineer • 2007 - 2010

My first three engineering jobs — web applications in PHP and Java, JavaScript front-ends and MySQL.

A selection of projects — the complete history, including those under Notable Achievements, is at <https://gil.solutions>

Education

Pontifícia Universidade Católica de São Paulo (PUC-SP) • Bachelor, Computer Science • 2008 - 2011